BYUNG HYUN LEE

CONTACT INFORMATION

Affiliation: Department of Electrical and Computer Engineering, Seoul National University

Email Website Google Scholar LinkedIn

RESEARCH INTERESTS

My research focuses on **selective and efficient post-training of foundation models**, enabling models to acquire new capabilities, remove unwanted knowledge, and adapt their behaviors while preserving existing performance. My work spans **continual learning, machine unlearning, and reinforcement learning**, with experience across visual, generative, and multimodal foundation models. Recently, I am interested in **multimodal LLMs, agentic AI, and vision-language-action models**, and in extending them to **domain-specialized AI**.

EDUCATION

- **Seoul National University (SNU)** *Mar 2021 - Feb 2027 (expected)*
  Combined M.S./Ph.D Program in Electrical and Computer Engineering (ECE)
- **Ulsan National Institute of Science and Technology (UNIST)** *Mar 2015 - Aug 2018*
  B.S. in Electrical Engineering, GPA: 4.04/4.30

RESEARCH ADVISOR

- **Se Young Chun:** Professor, Department of Electrical and Computer Engineering (ECE), SNU

PUBLICATIONS (Top AI Conference Papers)

- Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs
  H. Seo*, **B. H. Lee***, M. Kim*, D. Mah, J. Lee, S. Y. Chun
  Conference on Neural Information Processing Systems (NeurIPS, Spotlight), 2026
- Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models
  H. Seo*, **B. H. Lee***, J. Cho, S. Lim, S. Y. Chun
  Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis
  B. H. Lee, W. Jeong, W. Han, K. Lee, S. Y. Chun
  International Conference on Computer Vision (ICCV), 2025
- Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation
  B. H. Lee*, S. Lim*, S. Y. Chun
  Conference on Computer Vision and Pattern Recognition (CVPR), 2025
- Concept pinpoint eraser for text-to-image diffusion models via residual attention gate
  B. H. Lee*, S. Lim*, S. Lee, D. U. Kang, S. Y. Chun
  International Conference on Learning Representations (ICLR), 2025
- Doubly perturbed task free continual learning
  B. H. Lee, M. Oh, S. Y. Chun
  Proceedings of the AAAI Conference on Artificial Intelligence (AAAI, Oral), 2024

- Online Continual Learning on Hierarchical Label Expansion
  B. H. Lee*, O. Jung*, J. Choi, S. Y. Chun
  International Conference on Computer Vision (ICCV), 2023

- All-in-one image restoration for unknown degradations using adaptive discriminative filters for specific degradations
  D. Park, **B. H. Lee**, S. Y. Chun
  Conference on Computer Vision and Pattern Recognition (CVPR), 2023

PUBLICATIONS (Journal Papers)

- Continual Test-Time Adaptation for Robust Remote Photoplethysmography Estimation
  H. Lee, H. Lee, **B. H. Lee**, S. Y. Chun
  IEEE Access, 2025

- Towards accelerating model parallelism in distributed deep learning systems
  H. Choi*, **B. H. Lee***, S. Y. Chun, J. Lee
  PLOS One, 2023

TECHNICAL REPORTS / MANUSCRIPTS UNDER REVIEW

- Hyperbolic Vision-Language Model for Pathological Patch-Slide Representation
  B. H. Lee*, J. H. Jang*, Y. Kim, H. Kim, S. Y. Jeon, J. Cho, J. Lee, H. Seo, H. Bae, S. Y. Chun
  ICLR Submitted, 2027

- Multi-student On-Policy Self-Distillation with Alternating Reinforcement Learning for Robust Vision-Language-Action Models
  H. Kim*, **B. H. Lee***, J. Cho, J. Park, Y. Seo, Y. Hwang, D. Shin, S. Y. Chun
  ICLR Submitted, 2027

- S^3-OPSD: Sparse, Stable and Speedy On-Policy Self-Distillation
  H. Seo*, **B. H. Lee***, M. Oh, S. Y. Chun
  ICLR Submitted, 2027

- RePath: Zero-Shot Whole-Slide Tumor Segmentation via Intra-Slide Relational Affinity
  D. U. Kang*, S. Lim*, J. Cho, **B. H. Lee**, K. Lee, S. Y. Chun
  ICLR Submitted, 2027

- Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis
  H. Seo*, H. Bae*, **B. H. Lee***, J. Cho, J. Lee, J. H. Jang, H. Kim, S. Y. Jeon, Y. Kim, S. Y. Chun
  2026

- HyperCLOVA X 8B Omni
  HyperCLOVA X Team in NAVER Cloud (including **B. H. Lee**)
  2026

- HyperCLOVA X 32B Think
  HyperCLOVA X Team in NAVER Cloud (including **B. H. Lee**)
  2026

- Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation
  H. Seo*, J. Bang*, H. Lee*, J. Lee, **B. H. Lee**, S. Y. Chun
  arXiv, 2025

PUBLICATIONS (Short, Workshop, or Domestic Papers)

- Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models
  K. R. Lee*, K. H. Lee*, S. Hong*, **B. H. Lee**, S. Y. Chun
  CVPR Workshop on Machine Unlearning for Computer Vision, 2026
- Selective Concept Erasing for Safe Diffusion Models
  B. H. Lee, S. Lim, S. Y. Chun
  Korea Signal Processing Conference (Best Poster Presentation Award), 2024
- Expert classifier ensemble based post-processing correction for unbiased scene graph generation
  S. Lee, **B. H. Lee**, S. Y. Chun
  Workshop on Image Processing and Image Understanding (IPIU), 2024
- Efficient and accurate quantized image super-resolution on mobile npus, mobile ai & aim 2022 challenge: report
  A. Ignatov et al. (including **B. H. Lee**)
  European Conference on Computer Vision Workshops (ECCVW), 2022
- Efficient single-image depth estimation on mobile devices, mobile AI & AIM 2022 challenge: report
  A. Ignatov et al. (including **B. H. Lee**)
  European Conference on Computer Vision Workshops (ECCVW), 2022
- Uncertainty-based dual domain low-dose X-ray CT reconstruction
  S. Lee, D. U. Kang, **B. H. Lee**, S. Y. Chun
  Korean Signal Processing Conference, 2022
- Empirically Accelerating Scaled Gradient Projection Using Deep Neural Network for Inverse Problems in Image Processing
  B. H. Lee, S. Y. Chun
  International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2021

PATENTS

- **All-in-one image quality improvement model providing method performing image quality restoration for multiple image quality degradation factors**
  S. Y. Chun, D. Park, **B. H. Lee**
  U.S. Patent, Registered, 2026 (Filed, 2023)
- **Method for providing all-in-one image quality improvement model that performs image quality restoration for multiple image quality inhibitors**
  S. Y. Chun, D. Park, **B. H. Lee**
  Korea Patent, Registered, 2025 (Filed, 2023)

AWARDS AND HONORS

- **Compute Transparency Champion Award** *2026*
  CVPR 2026 (for "Erasing Thousands of Concepts")
- **Outstanding Reviewer (Top 5%)** *2026*
  Conference on Computer Vision and Pattern Recognition (CVPR)
- **Apple Scholars in AI/ML PhD Fellowship Program** *2025*
  Institutional Nominee
- **Qualcomm Innovations Fellowship Korea** *2025*
  Best paper finalist
- **2nd Place in Report Generation in Pathology using Pan-Asia Gigapixel WSIs** *2025*
  Challenge by International Conference on MICCAI (Served as the team leader, 26 teams participated)
- **Yulchon AI Star Scholarships** (₩8,000,000) *2025*
  Youlchon Foundation & SNU AI Institute

PROJECTS

- **Uncertainty-Aware Bidirectional Interaction between World Models and Vision-Language-Action Models** *Jul 2026 - Present*
  AI Research Computing Support Program, Korea Electronics Technology Institute (KETI), Republic of Korea
- **Hyperbolic Pathology Foundation Models for Gigapixel Report Generation** *Jan 2026 - Jun 2026*
  AI Computing Resource (GPU) Support Program, MSIT & NIPA, Republic of Korea
- **AI for Human-Machine Teaming** *Aug 2025 - Present*
  United States Air Force & IITP, Republic of Korea
- **Industrial Defect Segmentation** *Dec 2024 - Mar 2025*
  Samsung Device Solution (DS)
- **AI System for Gigapixel Histopathological WSI Analysis** *Aug 2023 - Dec 2024*
  Ministry of Health and Welfare, Republic of Korea
- **Relations and Affordance Aware Robot Grasping** *May 2022 - May 2023*
  Samsung Research, Republic of Korea

MEDIA COVERAGE

- **Popular Science Korea** *Jun 2026*
  Featured article on "Erasing Thousands of Concepts" (CVPR 2026), which erases 2,072 concepts from a text-to-image diffusion model

EXPERIENCES

- **National AI Foundation Model Project in South Korea**, NAVER Cloud *Oct 2025 - Feb 2026*
  Visiting Student Researcher
- **Biomedical Medical Image Processing Lab**, UNIST *May 2020 - Feb 2021*
  Researcher (Advisor: Prof. Se Young Chun)
- **Biomedical Medical Image Processing Lab**, UNIST *July 2017 - Aug 2018*
  Student Intership (Advisor: Prof. Se Young Chun)

PRESENTATIONS

- **Invited Poster Presentation on Selective Knowledge Updates for Deep Visual Models**
  Samsung AI Forum (SAIF) 2025, Samsung Electronics, Yongin, Korea, Sep 2025
- **Continual Learning and Its Applications in Magnetic Resonance Imaging**
  Advanced neuroimaging and AI workshop, SNU, 2024

ACADEMIC SERVICES

- **Reviewer**: NeurIPS, CVPR, ECCV, AAAI, IEEE Transactions on Image Processing (TIP), Pattern Recognition

EXTRACURRICULAR EXPERIENCES

- **Military Service, Republic of Korea Army** *Sep 2018 - Apr 2020*
  Discharged as Sergeant
- **Pinocchio (Robot Club)**, UNIST *Mar 2015 - Aug 2018*
  Education Director